

MATH356 (Honours Probability) Notes

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Table of Contents

1	September 1: Introduction to Probability Spaces	2
1.1	Motivation	2
1.2	Introduction to Probability Spaces	3
2	Uniform Sampling, Infinite Sample Space, Consequences	3
2.1	Uniform Sampling	3
2.2	Infinite Probability Spaces	4
2.3	Probability Consequences	4
3	September 8th: Inclusion/Exclusion Principles, Random Variables	6
3.1	Inclusion/Exclusion Principles	6
3.2	Random Variables	8
4	September 10th: Conditional Probability and Independent Events	8
4.1	Conditional Probability	8

1 September 1: Introduction to Probability Spaces

1.1 Motivation

Consider a fair six-sided die. Let D be the face that shows after a roll. We call D a **random variable**.

Suppose we roll a 2, i.e. $D = 2$. Then the probability of rolling this 2, denoted $\mathbb{P}(D = 2)$ is $1/6$ as we know very well (in fact $\mathbb{P}(D = k) = 1/6$ for any $k \in \{1, \dots, 6\}$).

Suppose now we roll the same die twice, with the face showing after the first roll denoted D_1 and the second roll D_2 . We ask ourselves the probability of first rolling a 2, then a 3 (i.e. $\mathbb{P}(D_1 = 2, D_2 = 3)$). Again, as we know very well, this probability is $1/36$.

Let us now consider two unfair six-sided dice. The first die has $\mathbb{P}(D = 4) = \mathbb{P}(D = 5) = \mathbb{P}(D = 6) = 2/9$ and $\mathbb{P}(D = 1) = \mathbb{P}(D = 2) = \mathbb{P}(D = 3) = 1/9$, while the second die has $\mathbb{P}(D = 4) = \mathbb{P}(D = 5) = \mathbb{P}(D = 6) = 1/9$ and $\mathbb{P}(D = 1) = \mathbb{P}(D = 2) = \mathbb{P}(D = 3) = 2/9$.

Let us blindly choose one of these unfair dice and roll it twice. We ask, what is $\mathbb{P}(D = 1)$? We don't know which die we chose, but we know that if we chose the first die, the probability of rolling a 1 is $1/9$, while if we chose the second die, the probability of rolling a 1 is $2/9$. We have a $1/2$ probability of choosing either die, so by laws of conditional probability and independent events (we will see these later), we have:

$$\mathbb{P}(D = 1) = \frac{1}{2} \cdot \frac{2}{9} + \frac{1}{2} \cdot \frac{1}{9} = \frac{1}{6}$$

In fact, as with the fair die, $\mathbb{P}(D = k) = 1/6$ for any $k \in \{1, \dots, 6\}$.

Example 1.1.1. Consider the same two unfair dice as above. What is $\mathbb{P}(D_1 = 1, D_2 = 1)$? In other words, what is the probability that both dice show a 1?

Solution. Suppose we chose the first unfair die (the probability of doing so is $1/2$). Then the probability of rolling a 1 twice is $1/9 \cdot 1/9 = 1/81$. If we chose the second die (again, the probability of doing so is $1/2$), then the probability of rolling a 1 twice is $2/9 \cdot 2/9 = 4/81$. Since we have a $1/2$ probability of choosing either die, we have that:

$$\mathbb{P}(D_1 = 1, D_2 = 1) = \left(\frac{1}{2} \cdot \frac{1}{9} \cdot \frac{1}{9} \right) \left(\frac{1}{2} \cdot \frac{2}{9} \cdot \frac{2}{9} \right) = \frac{5}{162} \quad \checkmark$$

This example may have been difficult to follow and that is okay. Many concepts such as “independent events” or “conditional probability” may be novel. The goal of this course is to learn what those topics are, among others:

- What is an event?
- What is a random variable?
- How can we model the outcome of an experiment?
- And many more!

1.2 Introduction to Probability Spaces

Definition 1.2.2

A **probability space** is a triple $(\Omega, \mathcal{F}, \mathbb{P})$, where Ω is the **sample space**, $\mathcal{F}$ is all subsets of Ω and $\mathbb{P}$ is the **probability measure**.

For any probability space, $\mathbb{P}$ must satisfy the **Kolmogorov Axioms**, which are:

1. If $A \in \mathcal{F}$, then $\mathbb{P}(A) \in [0, 1]$.
2. $\mathbb{P}(\emptyset) = 0$ and $\mathbb{P}(\Omega) = 1$.
3. If $(A_n)_{n \geq 1}$ are disjoint events, then $\mathbb{P}(\bigcup_{n \geq 1} A_n) = \sum_{n \geq 1} \mathbb{P}(A_n)$.

We call the elements in $\mathcal{F}$ **events**, such as rolling a 1, or flipping heads on a coin.

$\mathbb{P}$ assigns probabilities to events, with $\mathbb{P}: \mathcal{F} \rightarrow [0, 1]$.

Example 1.2.3. Let us revisit the fair six-sided die. Here we define $\Omega := \{1, 2, 3, 4, 5, 6\}$, $\mathcal{F} := 2^\Omega$ (the power set, or set of all subsets of Ω), and $\mathbb{P}(E) := \frac{|E|}{6}$.

If we roll the fair die twice, then now $\Omega := [6] \times [6]$, $\mathcal{F} := 2^\Omega$ and $\mathbb{P}(E) := \frac{|E|}{36}$.

Example 1.2.4. Model tossing a fair coin 3 times. What about infinitely many times? Can you model the unfair dice from before?

Solution. Left as an exercise (for next class). ✓

2 Uniform Sampling, Infinite Sample Space, Consequences

2.1 Uniform Sampling

Example 2.1.1. Suppose we choose three numbers from 1 to 50 at random. When we choose these numbers, we ask ourselves:

- Do we sample with or without replacement?
- Does the order matter?

Different answers to these questions will give different probabilities.

Sampling with replacement and ordered. Here, $\Omega := \{(a, b, c) : a, b, c \in [50]\}$ and we calculate $\mathbb{P}\{(a, b, c)\} = 1/50^3$.

Sampling without replacement and ordered. Here $\Omega := \{(a, b, c) : a, b, c \in [50] \text{ unique}\}$ and we calculate $\mathbb{P}\{(a, b, c)\} = 1/(50)_3$ ($(50)_3 := 50 \cdot 49 \cdot 48$).

Sampling without replacement and without order. Since order is irrelevant, $\Omega := \{S \subseteq [50] : |S| = 3\}$ and we calculate $\mathbb{P}\{\{a, b, c\}\} = 1/\binom{50}{3} = \frac{3!}{(50)_3}$.

Sampling with replacement and without order. Left as an exercise.

Example 2.1.2 (Lotto 6/49). Suppose we draw six numbers at random from 1 to 49 without replacement. Here, $\Omega := \{(a, b, c, d, e, f) : a, b, c, d, e, f \in [49] \text{ and distinct}\}$ and $\mathbb{P}: \Omega \rightarrow \mathbb{R}$. For any event $E \in \mathcal{F}$, $\mathbb{P}\{E\} = \frac{|E|}{|\Omega|}$. What is:

1. $\mathbb{P}\{(\text{odd}, \text{even}, \text{odd}, \text{even}, \text{odd}, \text{even})\}$

2. $\mathbb{P}\{\{3 \text{ odd}, 3 \text{ even}\}\}$

Solution. We first note that $|\Omega| = (49)_6$. We also note that there are 25 odd numbers and 24 even numbers in the sample space.

For (1), there are 25 odd numbers on the first draw, then 24 even numbers on the second draw, then 24 odd numbers on the third draw, then 23 even numbers on the fourth draw, then 23 odd numbers on the fifth draw, and finally 22 even numbers on the sixth draw. So we have that $|E| = (25)_3(24)_3$ and thus:

$$\mathbb{P}\{(\text{odd}, \text{even}, \text{odd}, \text{even}, \text{odd}, \text{even})\} = \frac{(25)_3(24)_3}{(49)_6}$$

For (2), there are $\binom{25}{3}$ ways to choose 3 odd numbers and $\binom{24}{3}$ ways to choose 3 even numbers from the space. Hence:

$$\mathbb{P}\{\{3 \text{ odd}, 3 \text{ even}\}\} = \frac{\binom{25}{3}\binom{24}{3}}{\binom{49}{6}} \quad \checkmark$$

2.2 Infinite Probability Spaces

Suppose we are throwing a dart at a uniformly random location on a dartboard of radius 1. We define $\Omega := B(0, 1)$ and for some region on the dartboard $R \subseteq \Omega$, $\mathbb{P}(R) = \frac{\text{Area}(R)}{\pi}$.

We claim that for any point $p \in \Omega$, $\mathbb{P}\{\{p\}\} = 0$. To prove this statement, we will first introduce an important lemma:

Lemma 2.2.3 (Monotonicity of Probability)

For any two events $A, B \in \mathcal{F}$ in a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that $A \subseteq B$, $\mathbb{P}(A) \leq \mathbb{P}(B)$.

Proof. Since $A \subseteq B$, we can write $B = A \cup (A^c \cap B)$. Since A and $A^c \cap B$ are disjoint, we can write $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(A^c \cap B)$, which is greater than or equal to $\mathbb{P}(A)$ since probabilities are non-negative. ■

Returning to the dartboard problem, let us fix $\epsilon > 0$ such that $B(p, \epsilon) \subseteq B(0, 1)$. The probability of hitting the ball around p , or $\mathbb{P}(p, \epsilon) = \epsilon^2$. Since $\{p\} \subseteq B(p, \epsilon)$, monotonicity gives $\mathbb{P}(\{p\}) \leq \epsilon^2$. However since $\epsilon > 0$ is arbitrarily small and $\mathbb{P}(\{p\}) \geq 0$, it follows that $\mathbb{P}(\{p\}) = 0$.

2.3 Probability Consequences

Here is an important identity for any event $A \subseteq \Omega$:

$$\mathbb{P}(A) + \mathbb{P}(A^c) = 1$$

This fact implies we can find the probability of an event by subtracting its complement from 1.

Kolmogorov's axioms tell us that if $(A_n)_{n \geq 1}$ are disjoint events, then $\mathbb{P}(\bigcup_{n \geq 1} A_n) = \sum_{n \geq 1} \mathbb{P}(A_n)$. However what do we do when the events are not necessarily disjoint?

Proposition 2.3.4 (Subadditivity of Probability)

Let $(A_i)_{i=1}^n$ be a sequence of events (not necessarily disjoint). Then $\mathbb{P}(\bigcup_{i=1}^n A_i) \leq \sum_{i=1}^n \mathbb{P}(A_i)$.

Proof. Let us "disjointify" the sets in $(A_i)_i$. To do so, we can write $A_1 = A_1$, $A_2 = A_1^c \cap A_2$, and so on until $A_n = A_1^c \cap A_2^c \cap \cdots \cap A_n$. With these disjoint sets, we can apply Kolmogorov's axiom of countable additivity:

$$\begin{aligned} \mathbb{P}(A_1 \cup A_2 \cup \cdots \cup A_n) &= \mathbb{P}(A_1 \cup (A_1^c \cap A_2) \cup \cdots \cup (A_1^c \cap A_2^c \cap \cdots \cap A_n)) \\ &= \mathbb{P}(A_1) + \mathbb{P}(A_1^c \cap A_2) + \cdots + \mathbb{P}(A_1^c \cap A_2^c \cap \cdots \cap A_n) \\ &\leq \mathbb{P}(A_1) + \mathbb{P}(A_2) + \cdots + \mathbb{P}(A_n) \\ &= \sum_{i=1}^n \mathbb{P}(A_i) \end{aligned} \quad \blacksquare$$

An important principle when solving probability problems is to break down a complicated event into simpler disjoint events whose union is equal to the complicated event, then compute the simple events and sum them up.

Example 2.3.5. Alice and Bob are taking turns rolling a fair six-sided die (Alice goes first, then Bob). Alice wins on a roll of $\{1, 2\}$ while Bob wins on a roll of $\{3, 4, 5, 6\}$. Find the probability $\mathbb{P}(\text{Alice rolls } \leq 4 \text{ times and wins})$.

Solution. Let $A_k :=$ the events where Alice wins on her k -th roll. Then $E := A_1 \cup A_2 \cup \cdots \cup A_n$. If the rolls are $a_1, b_1, a_2, b_2, \dots, a_k, b_k$, where a_i represents Alice's roll and b_i represents Bob's roll, then the probability of Alice winning on her k -th roll is:

$$\mathbb{P}(A_k) = \left(\frac{2}{3}\right) \left(\frac{1}{3}\right) \cdot \left(\frac{2}{3}\right) \left(\frac{1}{3}\right) \cdots \frac{1}{3} = \left(\frac{2}{9}\right)^{k-1} \cdot \frac{1}{3}$$

Hence the probability of Alice winning if she rolls less than four times is:

$$\begin{aligned} \mathbb{P}(\text{Alice rolls } \leq 4 \text{ times and wins}) &= \mathbb{P}(A_1) + \mathbb{P}(A_2) + \mathbb{P}(A_3) + \mathbb{P}(A_4) \\ &= \left(\frac{2}{9}\right)^0 \cdot \frac{1}{3} + \left(\frac{2}{9}\right)^1 \cdot \frac{1}{3} + \left(\frac{2}{9}\right)^2 \cdot \frac{1}{3} + \left(\frac{2}{9}\right)^3 \cdot \frac{1}{3} \\ &= \frac{1}{3} \sum_{k=1}^4 \left(\frac{2}{9}\right)^{k-1} \\ &= \frac{935}{2187} \end{aligned} \quad \checkmark$$

We were able to successfully solve this complicated problem by breaking it down into smaller, more manageable components.

3 September 8th: Inclusion/Exclusion Principles, Random Variables

3.1 Inclusion/Exclusion Principles

Fact 3.1.1

For any events A, B :

$$\mathbb{P}(A) = \mathbb{P}(A \cap B) + \mathbb{P}(A \cap B^c)$$

$$\mathbb{P}(B) = \mathbb{P}(B \cap A) + \mathbb{P}(B \cap A^c)$$

Using the these identities for $\mathbb{P}(A)$ and $\mathbb{P}(B)$, we have the following identity:

Fact 3.1.2

For any events, A, B :

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

Moreover, for any events A, B, C , we have:

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) + \mathbb{P}(B \cap C) - \mathbb{P}(A \cap B \cap C)$$

These identities can easily be proven either using a Venn diagram or by using the identities given in Fact 3.1.1.

And in fact, for any events $E_1, E_2, \dots, E_n$, we have the following identity:

Proposition 3.1.3 (Inclusion/Exclusion Formula)

For any events $E_1, E_2, \dots, E_n$:

$$\mathbb{P}\left(\bigcup_{k=1}^n E_k\right) = \sum_{k=1}^n (-1)^{k+1} \sum_{\substack{S \subseteq [n] \\ |S|=k}} \mathbb{P}\left(\bigcap_{i \in S} E_i\right)$$

Proof. In Fact 3.1.2 we already proved the identity holds for $n = 2$ and $n = 3$. We will use induction to show that the proposition holds for all $n \geq 2$.

Assume the proposition holds for all n . For $n + 1$, we have events $E_1, \dots, E_n, E_{n+1}$. Let $A := E_1, \dots, E_n$ and $B := E_{n+1}$. From the $n = 2$ case, we have $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$. By

induction, we have:

$$\begin{aligned}\mathbb{P}(A) &= \sum_{k=1}^n (-1)^{k+1} \sum_{\substack{S \subseteq [n] \\ |S|=k}} \mathbb{P}\left(\bigcap_{i \in S} E_i\right) \\ \mathbb{P}(B) &= \mathbb{P}(E_{n+1}) \\ A \cap B &= (E_1 \cup \dots \cup E_n) \cap E_{n+1} = (E_1 \cap E_{n+1}) \cup \dots \cup (E_n \cap E_{n+1}) \\ \mathbb{P}(A \cap B) &= \sum_{k=1}^n (-1)^{k+1} \sum_{\substack{S \subseteq [n] \\ |S|=k}} \mathbb{P}\left(\bigcap_{i \in S} E_i \cap E_{n+1}\right)\end{aligned}$$

So using the $n = 2$ case, we have:

$$\begin{aligned}\mathbb{P}(A \cup B) &= \sum_{k=1}^n (-1)^{k+1} \sum_{\substack{S \subseteq [n] \\ |S|=k}} \mathbb{P}\left(\bigcap_{i \in S} E_i\right) + \mathbb{P}(E_{n+1}) - \sum_{k=1}^n (-1)^{k+1} \sum_{\substack{S \subseteq [n] \\ |S|=k}} \mathbb{P}\left(\bigcap_{i \in S} E_i \cap E_{n+1}\right) \\ &= \text{AskatOfficeHours}\end{aligned}$$

■

Example 3.1.4. Suppose there are n individuals with n hats. Suppose that all n hats get mixed up. Each individual chooses a hat uniformly at random. What is the probability that at least one person gets their own hat back?

Solution. Let E_i be the event that person i gets their own hat. We want to find $\mathbb{P}\left(\bigcup_{i \in [n]} E_i\right)$. Using Proposition 3.1.3, we have:

$$\mathbb{P}\left(\bigcup_{i \in [n]} E_i\right) = \sum_{k=1}^n (-1)^{k+1} \sum_{\substack{S \subseteq [n] \\ |S|=k}} \mathbb{P}\left(\bigcap_{i \in S} E_i\right)$$

We know that for n hats, there are $n!$ possible results from this experiment, and moreover that there are $(n-k)!$ ways to distribute the hats among k individuals, hence $\mathbb{P}\left(\bigcap_{i \in S} E_i\right) = (n-k)!/n!$. Also note that there are $\binom{n}{k}$ terms in the inner summation. The equation simplifies to:

$$\begin{aligned}\mathbb{P}\left(\bigcup_{i \in [n]} E_i\right) &= \sum_{k=1}^n (-1)^{k+1} \binom{n}{k} \frac{(n-k)!}{n!} \\ &= \sum_{k=1}^n (-1)^{k+1} \frac{1}{k!} \\ &= 1 - \sum_{k=0}^n \frac{(-1)^k}{k!} \\ &\approx 1 - \frac{1}{e}\end{aligned}$$

✓

Proposition 3.1.5

If $A_1 \subseteq A_2 \subseteq \dots$ and $A_\infty = \bigcup_{k \geq 1} A_k$, then $\mathbb{P}(A_\infty) = \lim_{k \rightarrow \infty} \mathbb{P}(A_k)$.

Proof. Let us write $A_1 \dots A_n$ as disjoint sets. Write $B_1 = A_1$, $B_2 = A_1^c \cap A_2$, $\dots$, $B_n = A_{n-1}^c \cap A_n$ for $n \geq 2$. Then $(B_n)_{n \geq 1}$ are disjoint sets, and $\bigcup_{k \geq 1} B_k = \bigcup_{k \geq 1} A_k$, so we have:

$$\mathbb{P}(A_\infty) = \mathbb{P}\left(\bigcup_{k \geq 1} A_k\right) = \mathbb{P}\left(\bigcup_{k \geq 1} B_k\right) = \sum_{k=1}^{\infty} \mathbb{P}(B_k) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \mathbb{P}(B_k) = \lim_{n \rightarrow \infty} \mathbb{P}\left(\bigcup_{k=1}^n B_k\right) = \lim_{n \rightarrow \infty} \mathbb{P}(B_n)$$

■

3.2 Random Variables**Definition 3.2.6**

Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a **random variable** is a function $X: \Omega \rightarrow \mathbb{R}$.

For example, if we consider the sum of two fair die rolls, $\Omega := [6]^2$, and we can define $X: [6]^2 \rightarrow \mathbb{R}$, $X(i, j) = i + j$, where i is the value of the first roll and j is the value of the second. If we want to know $\mathbb{P}(X = 4)$, we can write $\mathbb{P}(\{i, j\} \in \Omega: X(i, j) = 4) = \mathbb{P}(\{i, j\} \in [6]^2: i + j = 4) = \frac{1}{12}$.

For a random variable $X: \Omega \rightarrow \mathbb{R}$, we write $x \in B$ to mean the set $\{\omega \in \Omega: X(\omega) \in B\} = X^{-1}(B)$.

Definition 3.2.7

The **distribution** of a random variable X is the collection of probabilities $\mathbb{P}(X \in B)$ for $B \subseteq \mathbb{R}$.

We could write the distribution as a function $2^{\mathbb{R}} \rightarrow [0, 1]$ with $B \mapsto \mathbb{P}(x \in B)$, but we don't need this much data. In fact, we could recover the distribution of X from the collection of probabilities $\{\mathbb{P}(X \leq b): b \in \mathbb{R}\}$.

There are three classes of random variables:

4 September 10th: Conditional Probability and Independent Events**4.1 Conditional Probability**